Wireshark Filter - Meaning - When to Use It

20 essential display filters for network traffic analysis

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How to use this sheet. These are Wireshark **display filters**. Type one in the filter bar at the top of Wireshark and press Enter. A green bar means the filter is valid, a red bar means there is a mistake. A display filter only hides packets from view, the capture itself is not changed. Symbols used: **==** equals, **&&** and, **||** or, **!** not, **contains** searches for text. The IP addresses are the lab addresses: Kali 192.168.56.101 and Ubuntu 192.168.56.102.

No.	Wireshark Filter	Meaning	When to Use It
A. Filter by host and port			
1	`ip.addr == 192.168.56.101`	Shows every packet where this IP address is the sender or the receiver.	When you want to study one machine only, for example the Kali VM, and hide all other traffic.
2	`ip.src == 192.168.56.101 && ip.dst == 192.168.56.102`	Shows only packets sent from the first IP address to the second one. The && means both conditions must be true, so replies are left out.	When you want to follow one direction of a conversation, for example what Kali sent to Ubuntu.
3	`tcp.port == 80`	Shows TCP packets that use port 80, the default web port, as source or destination.	To look at web traffic to the Apache server, such as the curl request.
B. TCP flags (handshake and scans)			
4	`tcp.flags.syn==1 && tcp.flags.ack==0`	Shows packets with the SYN flag on and the ACK flag off. These are the first packets that start a TCP connection.	To detect a port scan. A scanner sends a burst of these to many different ports in a short time. This is the filter used for the Nmap -sS screenshot.
5	`tcp.flags.syn==1 && tcp.flags.ack==1`	Shows SYN-ACK packets, the reply a host sends when a port is open and ready to connect.	To see which ports answered as open. During a scan, the ports that send SYN-ACK are the open ones.
6	`tcp.flags.reset==1`	Shows packets with the RST flag on. RST refuses or abruptly ends a connection.	To see closed ports answering a scan, or connections being refused.
7	`tcp.flags.fin==1`	Shows packets with the FIN flag on. FIN tells the other side that the sender has finished and wants to close the connection.	To check the normal teardown of a connection, and to spot FIN scans, where FIN packets arrive without any open connection.
C. Web and DNS traffic			
8	`http.request.method == "GET"`	Shows only HTTP requests that use the GET method, which asks a web server for a page or file.	To see which pages a client asked for, for example the curl request to the Apache server.
9	`http`	Shows all HTTP packets, both requests and responses. HTTP is not encrypted, so the content can be read.	To read a full web conversation, including the response codes the server sent back.

No.	Wireshark Filter	Meaning	When to Use It
10	`dns`	Shows all DNS packets. DNS turns a name into an IP address, so this covers the questions and the answers.	To see name lookups, for example the one made by the nslookup command, or to find which domains a computer is contacting.
11	`dns.qry.name contains "example"`	Shows DNS packets where the name being asked about contains the text in quotes. Replace example with the name you want.	To find lookups of one specific domain, for example a domain you find suspicious.
D. ICMP and ARP			
12	`icmp`	Shows all ICMP packets. ICMP is used by ping and for network error messages.	To find ping traffic and to see whether a host is sending ICMP errors.
13	`icmp.type == 8`	Shows ICMP echo requests, the packets sent by the ping command. Type 0 is the reply.	To see who is pinging whom. Many echo requests to different addresses in a row is a ping sweep, one way of finding live hosts.
14	`arp`	Shows all ARP packets. ARP asks who owns an IP address so a device can find the hardware address of another device on the same network.	To watch devices discovering each other, and to look for ARP spoofing, where one IP address is claimed by two different hardware addresses.
E. Encrypted traffic and Windows file sharing			
15	`tls.handshake.type == 1`	Shows the TLS Client Hello, the first message a client sends to start an encrypted HTTPS connection.	To find where HTTPS sessions begin, because the content after that is encrypted and cannot be read.
16	`smb || smb2`	Shows SMB and SMB2 packets, which Windows uses for file and printer sharing. The || means either condition can be true.	To check Windows file sharing activity. Unusual SMB traffic between computers can show an attacker moving from one machine to another.
F. Searching and analysing			
17	`frame contains "password"`	Searches the raw bytes of every packet for the text in quotes.	To find passwords sent as plain text in unencrypted traffic. Use it only on your own lab captures.
18	`tcp.analysis.retransmission`	Shows packets that Wireshark has marked as a re-send of data that seems to have been lost.	To find network problems such as packet loss or a slow connection.
19	`tcp.stream eq 0`	Shows all packets that belong to one TCP conversation. Wireshark numbers the conversations from 0.	To isolate a single connection, for example one curl request. Change the 0 to see other conversations. Right click a packet, then Follow, then TCP Stream, to read it.
20	`!(arp || dns)`	Hides ARP and DNS packets. The ! means not.	To remove background noise so the main traffic is easier to read.